

Progress Presentation: Drone Racing

Intelligent Planning + MPC + Reinforcement Learning

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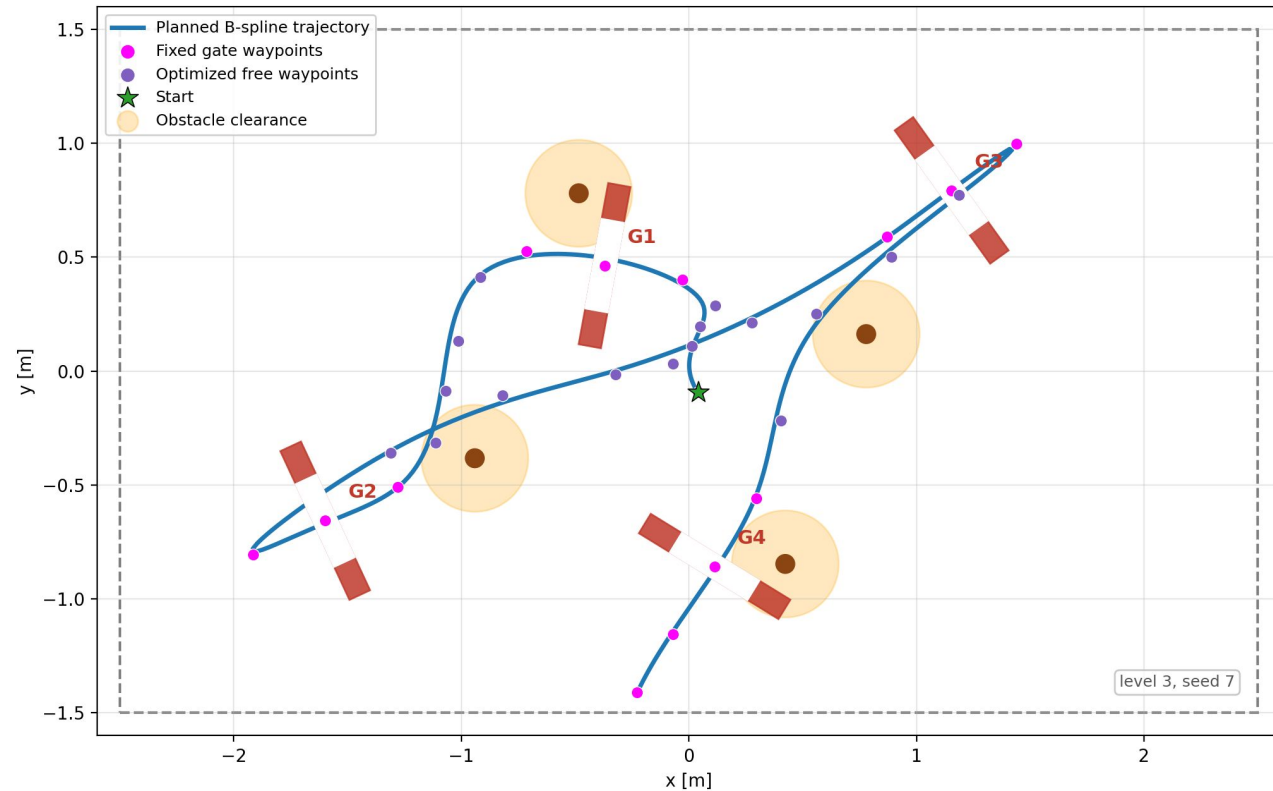
Learning Systems and Robotics Laboratory





Approach: Intelligent Path Planner

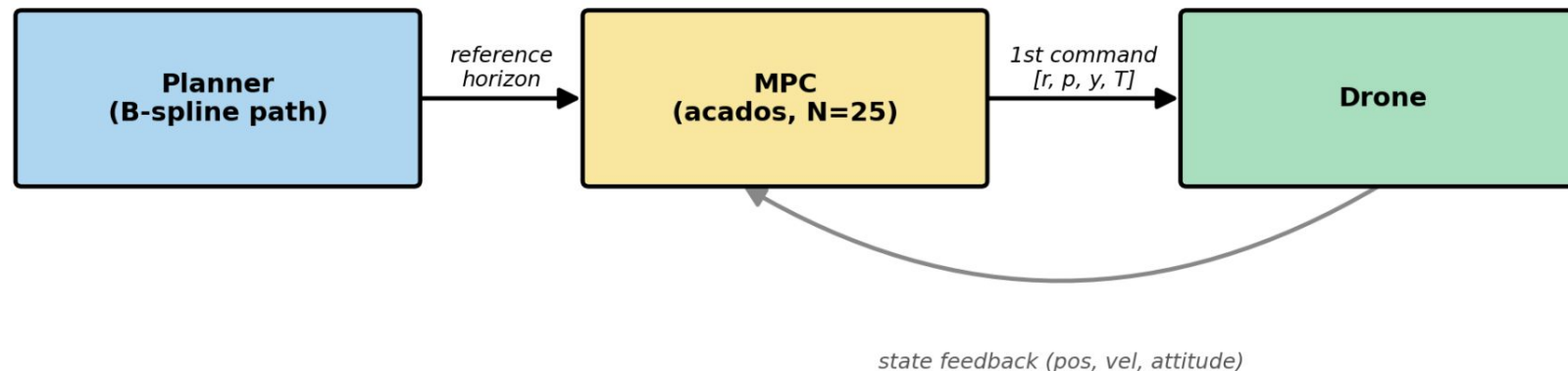
- 3 points per gate (before, center, after) → along the gate normal
- 4 freely movable points between gates → shift them to minimize a cost function
 - obstacle penalty
 - gate frame penalty
 - straight line deviation penalty
- cubic B-Spline
 - gate points: high weight
 - free points: low weight
- apply a speed profile
 - smooth acceleration
 - maximum speed (1.5 m/s)
 - sampled into uniform timesteps
- When a measured obstacle/gate position changes: Replan





Approach: Model Predictive Control (MPC):

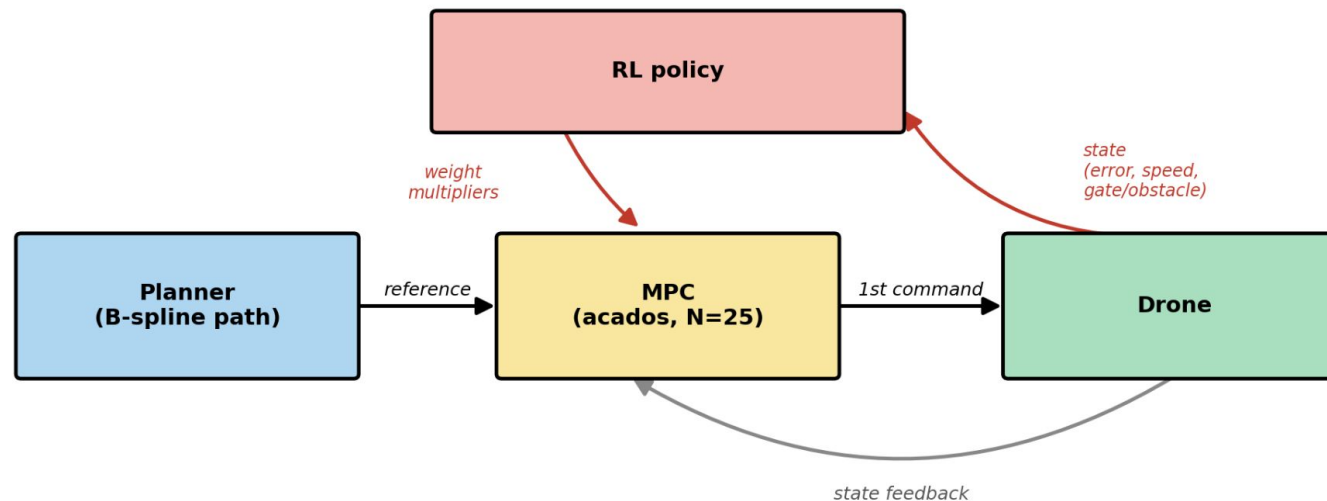
- Fetch the reference from the planner (horizon = 25)
- Measure current state
- Solve the MPC (acados) → optimal control sequence that:
 - tracks the reference position/velocity
 - respects physical limits (max tilt 30°, never under the ground, ...)
 - avoid obstacles/gate frames (soft constraint)
- Apply **only** the first command
- If solver fails: Use last good command





Approach: RL-adaptive MPC-tuning

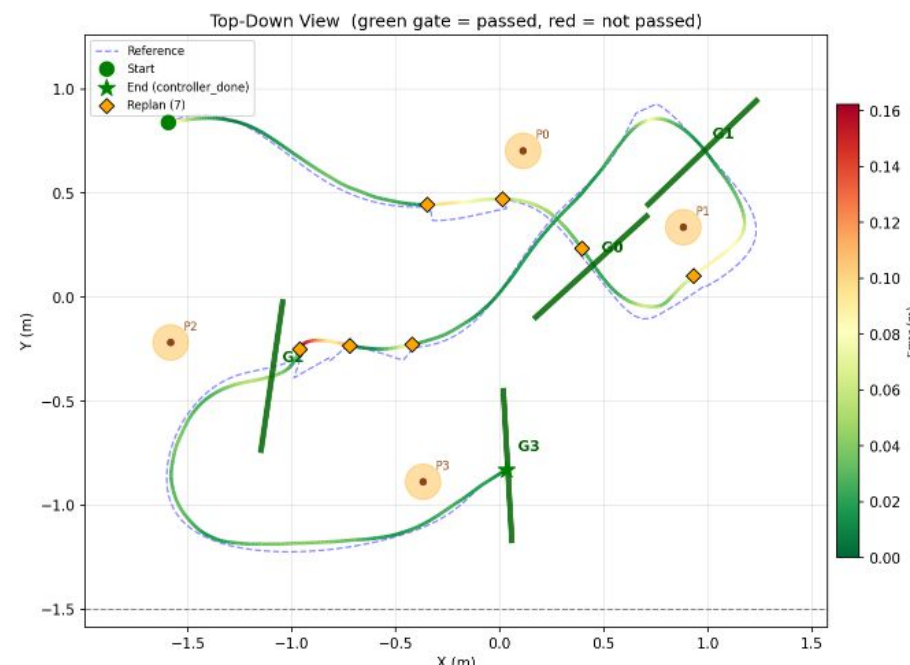
- Problem: MPC weights are static today → never optimal for every situation
- Idea: RL agent adapts the weights in real time to the drone's state
- Input (state): tracking error, current speed, distance/angle to gate & obstacle
- Output (action): multipliers on baseline weights (bounded → solver stays solvable)
- Division of roles: MPC flies → RL tunes
- Training: offline in sim, frozen policy during the race



Progress to Date



- A complex combined path planner does not work
 - In practice: higher optimization time, harder to tune, and still violate obstacle
 - Solution: switched to simple B-spline planner which optimizes the exact curve, simpler and more reliable
- Implemented the proposed planner and MPC Controller in Sim
 - level 0,1 works, though level 2 is still 60%
 - MPC is only roughly tuned
- Hardware Deployment Test
 - Environment issues (now fixed)
 - Can start testing in the next lab session





Project Milestones

1. Deploy the Path Planner and static MPC. (Jeffrey + Timo)
 - Achieve stable autonomous flight and successfully navigate the gate sequence without crashing level 2 on real hardware (Deadline: 17 June)
2. Train the RL weight-tuning agent (Timo), Refine the system based on Milestone 1 and Explore level 3 (Jeffrey).
 - RL agent converges in simulation; Base MPC successfully handles noisy state-estimation inputs on physical hardware.
3. Integrate the optimized RL policy (if proven stable) or the robust static baseline for final lab session.